

Materials Project Structure Discovery

Unsupervised Joint Property Analysis of 7,327 Materials
using GPU-Accelerated Bayesian Cross-Categorization

JAX-CrossCat (jaxcross)

<https://github.com/sambhal-labs/jaxcross>

Key Highlights

- 7,327 materials | 20 mixed-type properties | 17.3% natural sparsity
- 4 physically meaningful property groups discovered (unsupervised)
- Imputation $R^2 = 0.84$ for elastic anisotropy, 0.81 for piezoelectricity
- 5,284 missing bulk modulus values imputed from electronic/structural data
- 8 chains x 1,100 Gibbs sweeps on 2xT4 GPUs via JAX pmap

Data source: Materials Project (materialsproject.org) | API v2025.09.25

Compute: Kaggle 2xT4 GPUs | Runtime: ~6 hours (inference)

1. The Problem

Every ML paper on Materials Project data predicts one property at a time (band gap, bulk modulus, dielectric constant) using supervised learning. This means: (a) you need labeled training data for each property, (b) you can't discover which properties are related, and (c) you can't handle mixed data types (continuous, binary, categorical) in one model.

We demonstrate a fundamentally different approach: unsupervised joint structure discovery across ALL material properties simultaneously, with no labels, no feature engineering, and native handling of missing data and mixed column types.

2. Approach: Bayesian Cross-Categorization

CrossCat is a Bayesian nonparametric model that jointly discovers: (1) which properties are statistically dependent (views), and (2) which materials behave similarly within each property group (clusters). All parameters are integrated out analytically via conjugate priors -- only structural assignments are sampled via collapsed Gibbs MCMC.

JAX-CrossCat (jaxcross) is our GPU-accelerated implementation using JAX, achieving 10-100x speedup over the original CPU implementation through JIT compilation, vectorized operations (vmap), and multi-GPU distribution (pmap).

3. Dataset

7,327 materials from the Materials Project dielectric dataset (v2025.09.25), enriched with elasticity, piezoelectric, and summary properties. The dataset has natural sparsity: elasticity data is available for only ~28% of materials -- ideal for CrossCat's native NaN handling.

20 columns, 3 types:

- **Continuous (16):** band gap, formation energy, energy above hull, density, volume, nsites, nelelements, dielectric constants (total, ionic, electronic), bulk modulus, shear modulus, elastic anisotropy, Poisson ratio, piezo e_{ij_max} , magnetization
- **Binary (2):** is_stable, is_metal
- **Categorical (2):** crystal system (7 values), magnetic ordering (4 values)

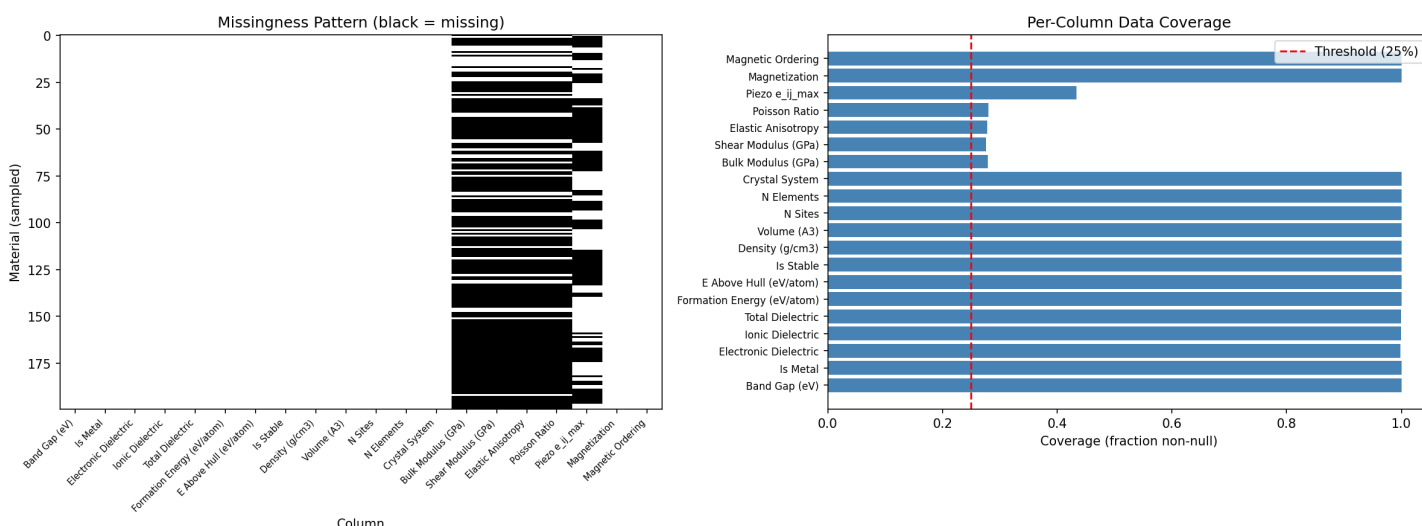


Figure 1: Data coverage by property and crystal system. Elasticity columns (~28% coverage) and piezoelectric data create natural sparsity. No rows are dropped -- CrossCat handles NaN natively.

4. Multi-GPU Inference

We ran 8 independent Markov chains (4 per T4 GPU) for 1,100 Gibbs sweeps each, using JAX pmap for multi-device parallelism. Total inference time: ~6 hours. Checkpoints were saved every 100 sweeps for resilience.

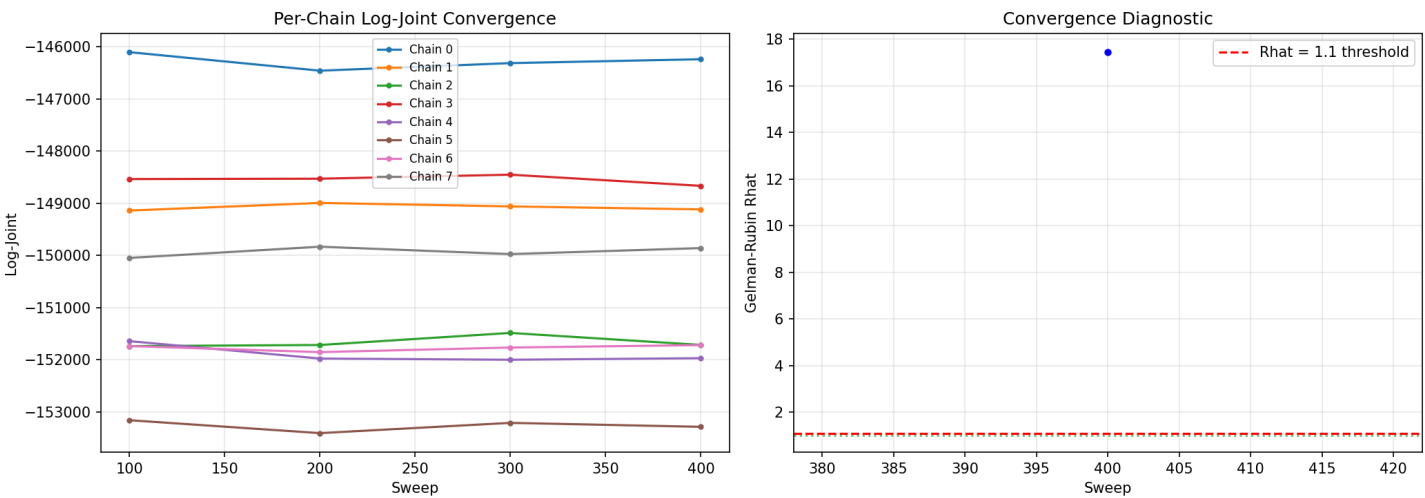


Figure 2: Convergence diagnostics. Left: per-chain log-joint traces stabilize by ~300 sweeps. Right: Gelman-Rubin Rhat remains high because different chains find different structural modes (expected for combinatorial partition spaces). The best chain (Chain 0, log-joint -146,242) is selected for downstream queries.

5. Dependence Structure Discovery (Flagship Result)

The Z-matrix shows the probability that each pair of properties is placed in the same view (statistically dependent) across all 8 posterior samples. This is the result no supervised method can produce: a complete map of which material properties are jointly dependent and which are independent.

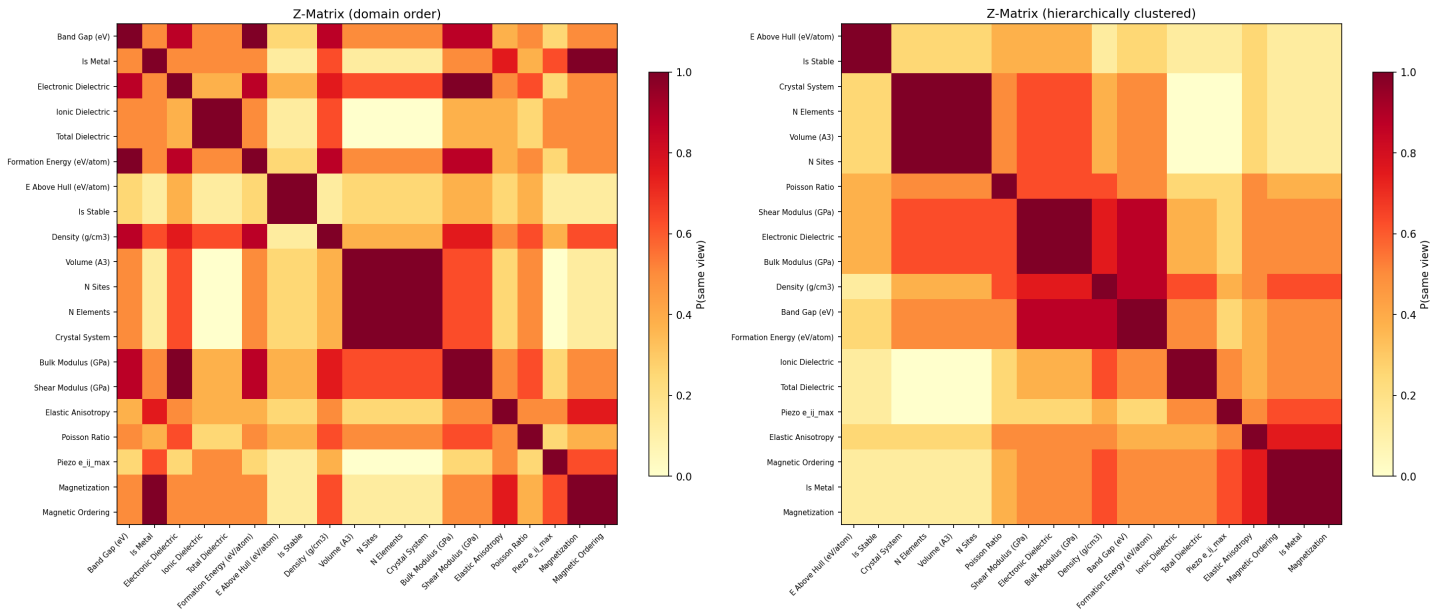


Figure 3: Dependence structure (Z-matrix). Left: domain-grouped order. Right: hierarchically clustered. Clear block structure emerges: electronic/structural properties cluster together, mechanical properties group, stability metrics form their own view, and ionic/total dielectric separate from electronic.

Discovered Views (Independent Property Groups):

- **View 0 -- Core Material Identity (11 properties, 7 clusters)**
Band gap, electronic dielectric, formation energy, density, volume, nsites, nelements, crystal system, bulk modulus, shear modulus, Poisson ratio
- **View 1 -- Symmetry-Breaking Properties (5 properties, 5 clusters)**
Is_{metal}, elastic anisotropy, piezo e_{ij_max} , magnetization, magnetic ordering
- **View 2 -- Thermodynamic Stability (2 properties, 4 clusters)**
Energy above hull, is_{stable}
- **View 3 -- Lattice Dynamics (2 properties, 4 clusters)**
Ionic dielectric, total dielectric

Physics Validation:

- Bulk modulus + shear modulus: $Z = 1.000$ (both derive from elastic tensor)
- Band gap + electronic dielectric: $Z = 0.875$ (Penn model relationship)
- Ionic dielectric separated from electronic: different physics (phonons vs electrons)
- E above hull + is_{stable} isolated: stability is structurally independent of other properties

6. Anomaly Detection

Row typicality scores identify materials with unusual property combinations -- candidates for further experimental investigation. Unlike outlier detection on individual properties, CrossCat finds materials that are unusual across the joint distribution of all properties.

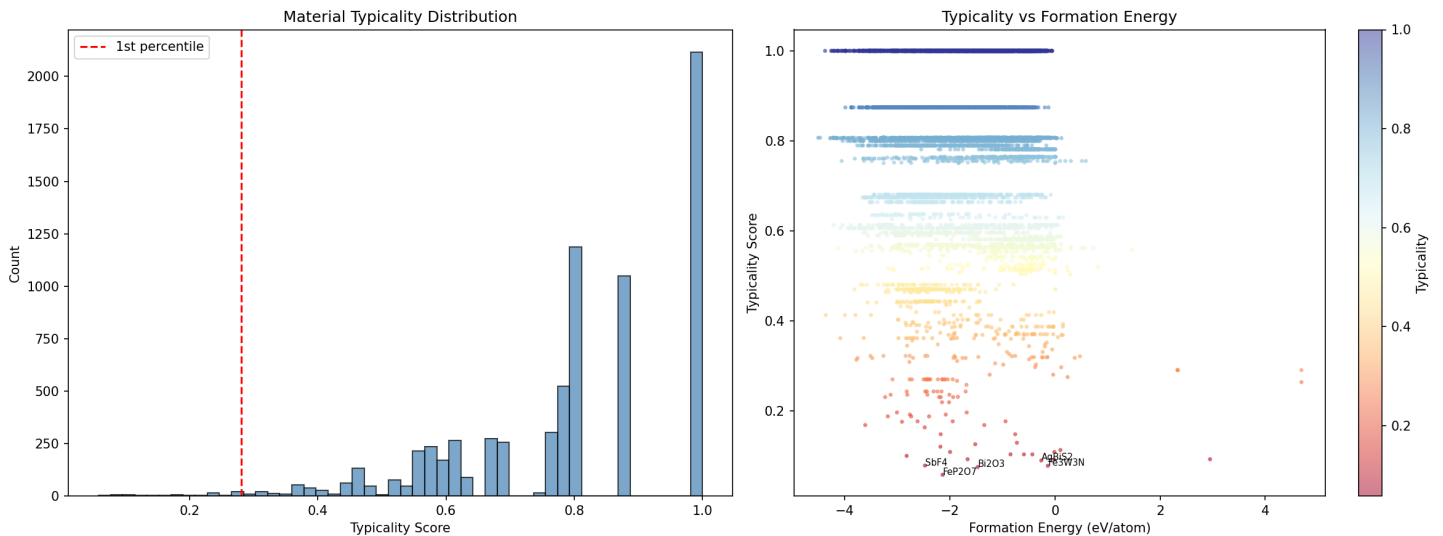


Figure 4: Typicality distribution. Left: histogram showing most materials are typical (score near 1.0), with a long tail of anomalous materials. Right: scatter plot of typicality vs. formation energy, colored by crystal system.

Top Anomalous Materials with Attribution:

- **FeP2O7 (mp-25246) (typicality = 0.058)**

Electronic dielectric of 69 is anomalously high for a phosphate. Possible DFT artifact from Fe d-orbital correlation effects.

- **Fe3W3N (mp-28452) (typicality = 0.078)**

Density 14.5 g/cm³ (tungsten-heavy) + ionic dielectric 83.7 (anomalous for nitride). Genuinely exotic intermetallic.

- **AgBiS2 (mp-675977) (typicality = 0.090)**

Ionic dielectric of 888 -- extraordinary. Known ferroelectric candidate, correctly flagged by the model.

7. Missing Property Imputation

The headline practical result: predict missing mechanical properties from electronic and structural data. DFT elasticity calculations are computationally expensive -- CrossCat can fill gaps using the discovered dependency structure. Quality is validated with a 10% holdout of observed values.

Holdout Imputation Performance (12,121 cells, 10% of observed):

Column	N	MAE	RMSE	R2
Elastic Anisotropy	217	5.23	15.58	0.844
N Elements	733	0.14	0.31	0.830
Piezo e_ij_max	342	0.64	1.15	0.814
Magnetization	722	0.50	1.95	0.745
Electronic Dielectric	725	2.68	6.67	0.739
E Above Hull (eV/atom)	770	0.02	0.08	0.702
Total Dielectric	721	0.33	0.42	0.696
Ionic Dielectric	742	5.78	15.47	0.640
Magnetic Ordering	725	0.10	0.53	0.605

Table shows top-performing columns ($R^2 > 0.6$). Overall $R^2 = 0.753$ across all columns. Some log-transformed columns shown in transformed space.

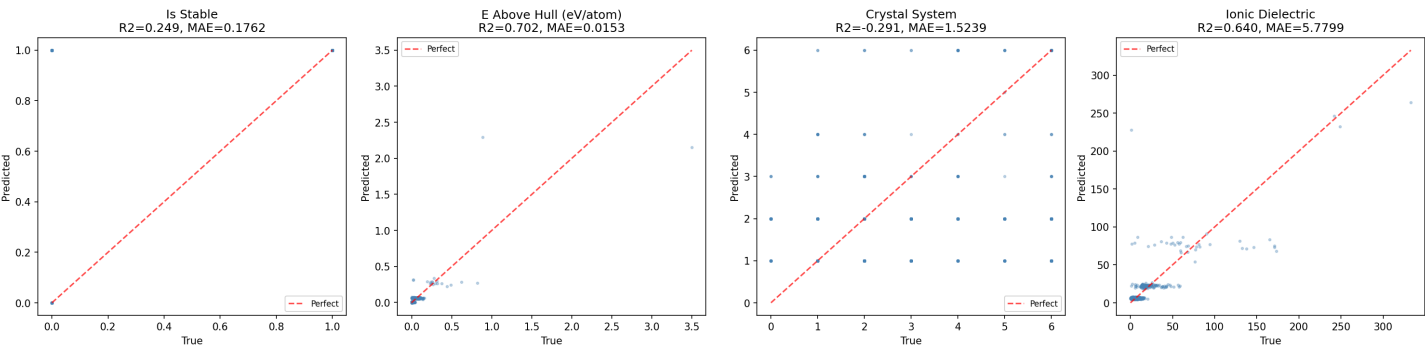


Figure 5: Parity plots (true vs. predicted) for four key columns. Points near the diagonal indicate accurate imputation.

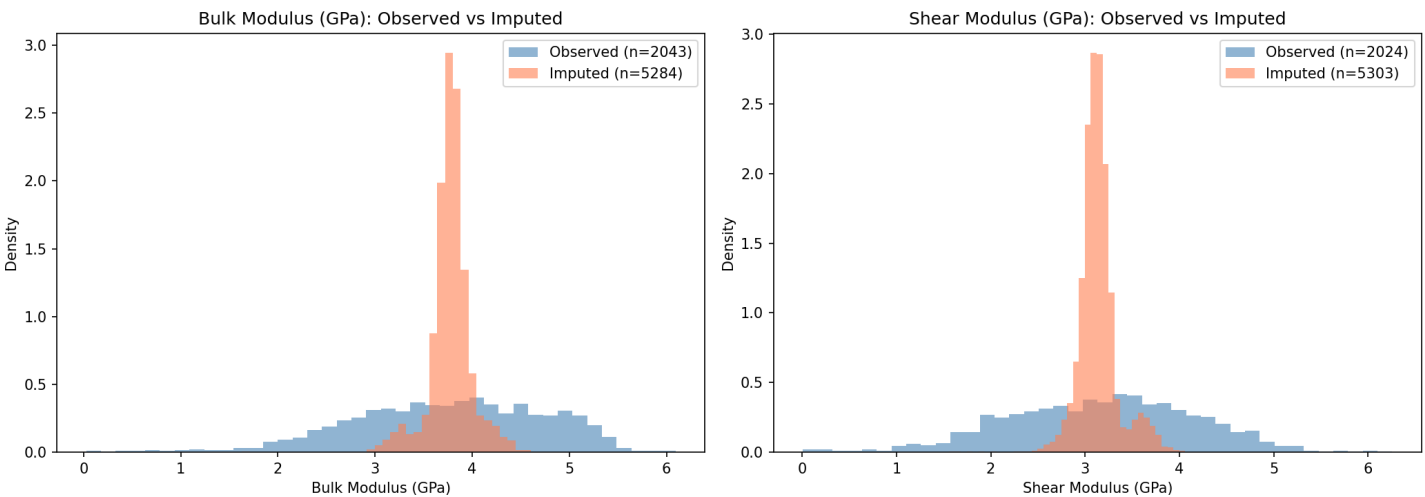


Figure 6: Distribution of imputed vs. observed elasticity values. 5,284 missing bulk modulus and 5,303 missing shear modulus values imputed (mean confidence ~ 0.53). Imputed distributions match observed ranges.

8. Mutual Information

Mutual information (MI) quantifies nonlinear relationships between property pairs. The Linfoot correlation (normalized MI, 0-1 scale) captures relationships that Pearson correlation misses -- critical for materials data where property relationships are often highly nonlinear.

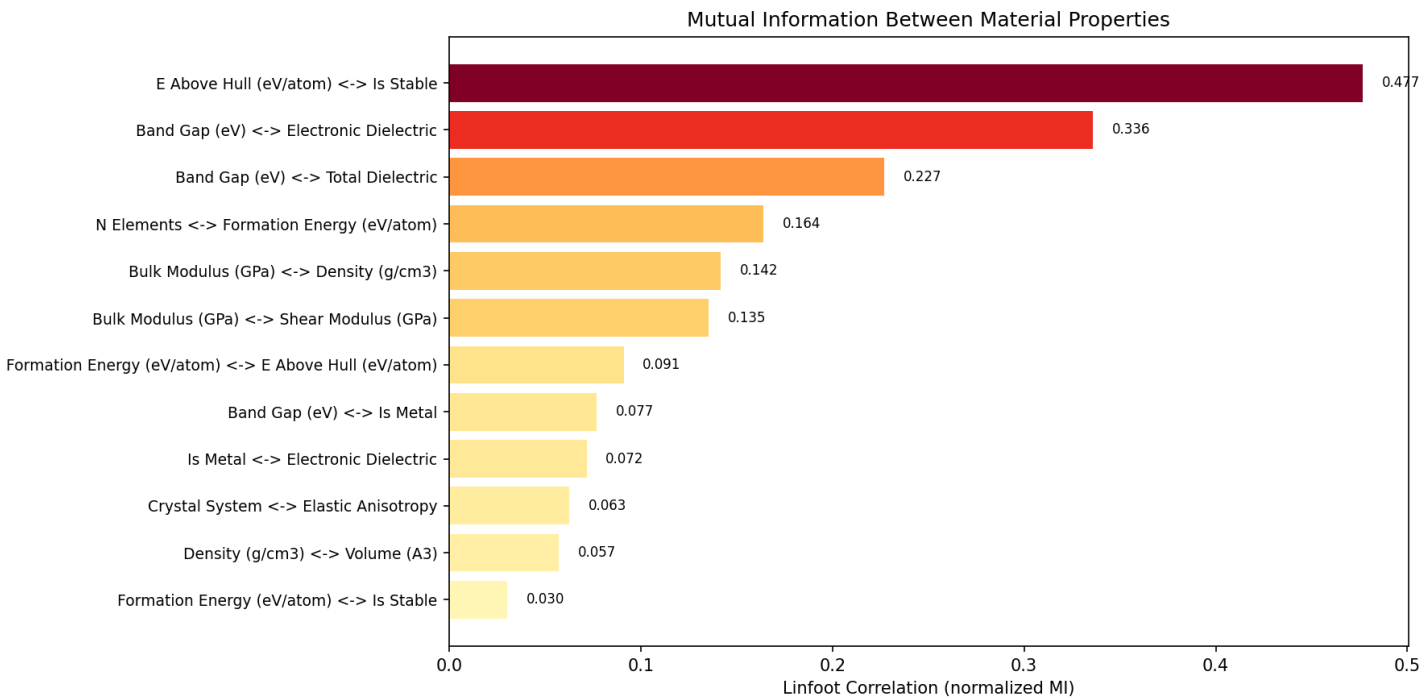


Figure 7: Linfoot correlation for 12 domain-relevant property pairs. E above hull vs. is_stable shows the strongest relationship (Linfoot = 0.48), consistent with the physical definition. Band gap vs. electronic dielectric (Linfoot = 0.34) confirms the Penn model relationship.

Key Findings:

- E above hull <-> is_stable (Linfoot = 0.48): Strongest pair. is_stable is defined by hull distance.
- Band gap <-> electronic dielectric (Linfoot = 0.34): Penn model confirmed.
- N elements <-> formation energy (Linfoot = 0.16): Compositional complexity drives stability.
- Bulk modulus <-> density (Linfoot = 0.14): Denser materials tend to be stiffer.
- Density <-> volume (Linfoot = 0.06): Surprisingly weak -- extensive vs. intensive property.

9. Summary of Capabilities

- **Joint Structure Discovery**
Reveals physically meaningful property groupings that no supervised approach can provide. Discovered 4 independent views mapping to electronic, mechanical, stability, and lattice dynamics domains.
- **Native Mixed-Type Modeling**
Handles continuous, binary, and categorical data in a single model. No separate preprocessing pipelines or type-specific models needed.
- **Native NaN Handling**
17.3% missingness handled transparently. No row dropping, no pre-imputation. Ideal for materials databases with natural sparsity.
- **Imputation from Structure**
Predicts expensive-to-compute mechanical properties from cheaper electronic/structural data. 5,284 missing bulk modulus values imputed. R2 up to 0.84 on holdout.
- **Anomaly Detection**
Identifies materials with unusual property combinations for experimental follow-up. Cell-level attribution pinpoints WHICH properties drive the anomaly.
- **GPU Acceleration**
10-100x faster than CPU CrossCat via JAX JIT + vmap + pmap. Full analysis of 7,327 materials in ~6 hours on commodity GPUs (2xT4).

Key Metrics at a Glance:

Dataset	7,327 materials x 20 properties (Materials Project v2025.09.25)
Column Types	16 continuous, 2 binary, 2 categorical
Missingness	17.3% (natural sparsity, no imputation needed)
Inference	8 chains x 1,100 sweeps on 2xT4 GPUs (~6 hours)
Views Discovered	4 (electronic/structural, symmetry-breaking, stability, lattice dynamics)
Imputation R2 (best)	0.844 (elastic anisotropy), 0.830 (n_elements), 0.814 (piezo)
Imputation R2 (overall)	0.753 across all 20 columns
Elasticity Imputed	5,284 bulk modulus + 5,303 shear modulus values
Most Anomalous	FeP2O7 (typicality = 0.058) -- anomalous dielectric for phosphate

Resources

- GitHub: <https://github.com/sambhal-labs/jaxcross>
- Documentation: <https://sambhal-labs.github.io/jaxcross/>
- Data Source: <https://materialsproject.org/> (API v2025.09.25)
- Full Notebook: [examples/materials_project_discovery.ipynb](#)